

Investigation:

Before writing any code, I read through the CSV carefully to understand what could be trusted and what needed inferring to inform my build decisions.

The clearest finding was that priority almost always follows from category and subcategory: tickets in the same subcategory cluster around the same difficulty level. Resolution time showed a linear relationship with priority but had enough outliers that I couldn't rely on it. A taxonomy is created to identify the base priority given the category and subcategory. The priority can be bumped up if conditions like urgency signals ("legal", "formal complaint", "completely unusable") appear in the description, or if the same customer has two or more prior tickets on the same issue.

I also spotted a column shift on some rows where the ticket description had landed in the `satisfaction_score` column because `resolution_time_hours` was blank, fixed on ingest by checking whether that field contains non-numeric text. Subcategory casing was also inconsistent in places (setup, Overage) and this was normalized for consistency.

For category matching, ticket descriptions and resolution notes contain very obvious domain keywords: "API", "hallucination", "webhook". I built a taxonomy of categories and subcategories with associated keywords, used RAKE to extract key phrases from each ticket description from the current ticket, and scored matches across the taxonomy. Highest score wins. This works well at this scale but will struggle when wording changes while meaning stays the same, at that point I'd swap in a BERT-based sequence classifier fine-tuned on labelled tickets.

For recurrence, ~10 tickets explicitly reference earlier ticket IDs in the ticket description. A regex on TKT-NNN plus a same-customer check catches these reliably. I considered TF-IDF cosine similarity on description vectors as a fallback but the dataset is too small for the threshold to be meaningful. For further improvement at scale, I would fine tune a BERT model, and then performing a pairwise sequence classification task from the ticket's description/resolution notes to capture if both sentences are reference to each other.

The human readable summary per ticket is intentionally concise: ticket, customer, suggested labels, linked tickets, status for the manager to take quick action.

From the file, it seems that agent assigned are blank for open/escalated tickets. Hence in the dashboard, I've included specific open/escalated tickets that does not have an agent, this is also ordered by their priority. Along that, with some general insights to the dashboard such as, satisfaction pattern, categories distribution, top-recurring customer hotspots, etc.